

3.1 Finding 1: Cross-Indicator Inconsistency

Tethered to a monolithic indicator paradigm, existing visual token pruning methods rest upon an assumption that has eluded systematic scrutiny: a single importance criterion suffices to separate indispensable tokens from expendable ones. Our investigation reveals that the indicators employed by existing methods naturally coalesce into two distinct families: *saliency*, which quantifies individual token importance (VisionZip [54], CDPPruner [58]), and *diversity*, which promotes collective representativeness of the retained set (DivPrune [3], SCOPE [11]).

As illustrated in Fig. 3, under an identical compression ratio, these two indicator families yield markedly divergent token selections. VisionZip, guided by ViT CLS attention, preferentially retains foreground tokens of high semantic saliency—a selection that remains invariant to the user’s query. CDPPruner augments this saliency signal with text–visual cosine similarity, redirecting retention toward query-relevant regions while its conditional DPP kernel enforces geometric spread among the selected tokens. By contrast, DivPrune pursues a fundamentally different objective: maximizing the minimum pairwise distance in the feature space, it scatters selections broadly across the image at the expense of fine-grained foreground detail. SCOPE mediates between these two poles, coupling CLS-based saliency with a submodular coverage term that curbs token clustering within narrow semantic neighborhoods.

This divergence arises because the four methods optimize along fundamentally *orthogonal indicator dimensions*:

- **Dimension A — Saliency** evaluates individual token importance and further decomposes into two sub-dimensions with distinct behaviors:
 - **A1 (Vision-Intrinsic):** “*Is this token intrinsically important in the image, regardless of the question?*”—relying solely on vision-encoder signals (e.g., CLS attention in VisionZip and SCOPE), producing a fixed ranking for a given image.
 - **A2 (Task Relevance):** “*Is this token relevant to the user’s current query?*”—requiring text–visual interaction (e.g., CDPPruner computes cosine similarity between each visual token and the instruction embedding), yielding query-dependent rankings that differ across prompts.
- **Dimension B — Representativeness** evaluates whether the selected subset collectively represents the original token set, and further decomposes along two complementary spaces:
 - **B1 (Semantic Diversity):** “*Are the selected tokens semantically non-redundant in the embedding space?*”—enforced via set-theoretic objectives that penalize feature-space duplication (e.g., DivPrune’s MMDP maximizes the minimum pairwise distance; SCOPE’s submodular coverage maximizes the sum of nearest-neighbor similarities; CDPPruner’s conditional DPP maximizes the volume spanned by the selected feature vectors).
 - **B2 (Spatial Coverage):** “*Does the selected set preserve at least one representative token from every spatial region of the image?*”—enforced via geometric partitioning of the token grid (e.g., Nuwa [39] divides the grid into $M \times M$ non-overlapping blocks and selects top- n tokens per block; HoloV [44] allocates per-crop budgets proportional to holistic scores). Notably, B1 and B2 are not redundant: a semantically diverse subset may still concentrate in a single spatial quadrant, while a spatially uniform selection may retain redundant tokens from texturally homogeneous regions.

Crucially, these dimensions are not merely orthogonal—their failure modes are *complementary*. Optimizing saliency alone (Dimension A) causes the retained tokens to cluster around a few high-attention foreground regions, leaving the semantic space severely under-covered; SCOPE’s analysis quantifies this: under pure CLS-attention selection on LLaVA-1.5, the θ -coverage of the retained set falls *below that of random sampling* across all threshold values [11]. Optimizing semantic diversity alone (B1) yields the opposite pathology: DivPrune distributes tokens broadly in the embedding space but is agnostic to which regions are task-relevant, retaining background noise at the expense of fine-grained foreground details. Furthermore, even a semantically diverse subset (B1) may remain spatially skewed—tokens from texturally varied but co-located patches can achieve high pairwise distance in feature space while leaving entire image quadrants unrepresented, a systematic weakness for OCR, spatial reasoning, and referring expression comprehension [39].

These complementary failure modes reveal that no single indicator dimension suffices: saliency without representativeness discards context; representativeness without saliency discards detail; and semantic diversity without spatial awareness discards regions. This analysis motivates the design of a unified pruning framework that jointly optimizes across all four sub-dimensions (A1, A2, B1, B2), as we present in Sec. 4.